

# Problem 1: Fixing Buggy Code using super()

```
class Person:
    def __init__(self, name, id):
        self.name = name
        self.id = id
    def Display(self):
        print("Name:", self.name)
        print("ID:", self.id)

class Emp(Person):
    def __init__(self, name, id):
        super().__init__(name, id)

    def Print(self):
        print("Emp class called")
Emp_details = Emp("Shristi", 28)
Emp_details.Display()
Emp_details.Print()
```

Name: Shristi  
ID: 28  
Emp class called

# Problem 2: Anagram Check

```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")
str1 = str1.replace(" ", "").lower()
str2 = str2.replace(" ", "").lower()
if sorted(str1) == sorted(str2):
    print("The strings are Anagrams.")
else:
    print("The strings are not Anagrams.")
```

Enter first string: silent  
Enter second string: listen  
The strings are Anagrams.

# Problem 3: Armstrong Number

```
num = int(input("Enter a number: "))
original_num = num
digits = len(str(num))
sum = 0
while num > 0:
    digit = num % 10
    sum = sum + digit ** digits
    num = num // 10
if sum == original_num:
    print(original_num, "is an Armstrong number.")
else:
    print(original_num, "is not an Armstrong number.")
```

Enter a number: 153  
153 is an Armstrong number.

# Problem 4: Classes and Objects

```
class Vehicle:
    def __init__(self, max_speed, mileage):
        self.max_speed = max_speed
        self.mileage = mileage
        self.color = "white"
        self.seating_capacity = 0
    def assign_seating_capacity(self, seating_capacity):
        self.seating_capacity = seating_capacity
    def display_properties(self):
        print("Maximum Speed:", self.max_speed, "km/h")
        print("Mileage:", self.mileage, "kmpl")
        print("Color:", self.color)
        print("Seating Capacity:", self.seating_capacity)
        print()
car1 = Vehicle(200, 20)
car1.assign_seating_capacity(5)
car2 = Vehicle(180, 25)
car2.assign_seating_capacity(4)
print("Car 1 Details:")
car1.display_properties()

print("Car 2 Details:")
car2.display_properties()
```

Car 1 Details:  
Maximum Speed: 200 km/h  
Mileage: 20 kmpl  
Color: white  
Seating Capacity: 5

Car 2 Details:  
Maximum Speed: 180 km/h  
Mileage: 25 kmpl  
Color: white  
Seating Capacity: 4

